

Deep Research Arena:

A Reachability-Grounded Benchmark Reveals Systematic URL Fabrication in Open-Source Deep Research Agents

Anonymous

Abstract

Open-source Deep-Research (DR) agent frameworks are typically evaluated with short cross-site tasks and LLM-as-a-judge rubrics. We show that this protocol rewards *fluent fabrication*: agents that cite plausibly-formatted but non-existent URLs can score at the top of the leaderboard. We introduce DR-ARENA-DEEP, a 10-topic benchmark in which each task requires an agent to cite at least 60 URLs spanning three dimensions (product / community / encyclopedic) of a reproducible sandbox, and a *truthfulness-first composite score* anchored to HTTP-level URL reachability. On DR-ARENA-DEEP five well-known open-source agents run with DeepSeek-V4-flash produce 0–100% URL reachability, whereas a state-of-the-art LLM judge (GPT-5-chat) marks most fabricated reports as 21/21 PASS. The truthfulness-first composite *inverts* the leaderboard: agents that the legacy metric ranks 1st–3rd are ranked 4th–6th when URL truth is enforced. We release the dataset, verifiers, and all agent traces.

1 Introduction

The last two years have produced a generation of open-source Deep-Research (DR) agent frameworks — GPT-Researcher, LangChain OPEN_DEEP_RESEARCH, HuggingFace smolagents, Stanford STORM, CAMEL-AI, DeerFlow, MindSearch, among others. Each positions itself on a leaderboard, typically of cross-site factoid or short-form retrieval benchmarks, with LLM-as-a-judge scoring.

This paper reports a negative result in that protocol. We built DR-ARENA-DEEP, an extension of our existing cross-site benchmark in which each task requires at least 120 URL visits during research and at least 60 distinct URL citations in the final report, spanning three orthogonal dimensions of a reproducible sandbox (Magento storefront, Postmill-style forum, and a Kiwix English-Wikipedia mirror). We then ran five popular open-source DR frameworks under the same DeepSeek-V4 backbone and GPT-5-chat judge — and found that three of the five *fabricate nearly every cited URL*. GPT-Researcher and LangChain OPEN_DEEP_RESEARCH produce reports in which **0 of 30 sampled URLs resolve on the sandbox**, yet the GPT-5-chat judge marks them 21/21 PASS. Meanwhile, STORM and CAMEL-AI, whose reports cite at 100% and 97% reachability respectively, sit at the bottom of the legacy composite because they are brief.

Our contributions:

1. **DR-Arena-Deep** (Section 3), a reproducible 10-topic cross-site sandbox benchmark with a documented task-generation protocol and machine-verifiable gold references.
2. A **truthfulness-first evaluation suite** (Section 4) consisting of (a) a URL-reachability verifier, (b) a URL-coverage verifier anchored to a curated pool, (c) a deterministic markdown spec check, and (d) a cross-family LLM-judge checklist. Composite is multiplicative in reachability, so any agent with near-zero URL truth scores near-zero composite.

3. An **empirical audit** (Section 6) of five popular OSS DR agents on DR-ARENA-DEEP with a DeepSeek-V4-flash backbone, demonstrating complete ranking inversion between the legacy and truthfulness-first composites.
4. Two named findings: **F5** — LLM judges (GPT-5-chat) cannot detect URL fabrication; **F6** — the currently-standard additive LLM-judge composite *systematically rewards* fluent fabrication over truthful brevity.

2 Related work

Deep Research benchmarks. DeepResearch Bench [Anonymous, 2024b] and BrowseComp-Plus [Anonymous, 2025a] evaluate research quality with short (3–20 citation) prompts on the open web. DRACO [Anonymous, 2024c] and the ByteDance Deep Research Survey [Research, 2025] both pair LLM judges with citation counts. None test URL reachability. WebArena [Zhou et al., 2024] pioneered the reproducible sandbox for web-agent tasks but uses goal-state matching rather than citation audit.

Fabrication detection. A body of work on LLM hallucination tests factual correctness [Anonymous, 2024d, Lin et al., 2022]. To our knowledge, no prior work evaluates DR agents on URL-level reachability. The closest work, CiteBench [Anonymous, 2024a], evaluates whether citations support claims but assumes the citations resolve.

Judge reliability. JudgeBench [Anonymous, 2025c] documents self-preference bias in LLM judges. Our F5 extends this to a judge *blind spot*: cross-family judges still cannot detect URL truth.

3 The DR-Arena-Deep benchmark

3.1 Design principles

DR-ARENA-DEEP tasks are constructed so that no agent can satisfy them by hallucinating fluent markdown. Each task requires: (C1) ≥ 120 sandbox URLs touched during research, (C2) ≥ 60 URLs cited as markdown links, (C3) citations spanning ≥ 3 distinct sandbox domains, (C4) every cited URL must return HTTP 200, (C5) $\geq 50\%$ of a curated must-cite pool appears in the report, (C6) the report is 3500–8000 words across ≥ 25 paragraphs, and (C7) the synthesis section surfaces ≥ 5 cross-source findings. Of these, only **C4 is not gameable by fluent fabrication**.

3.2 Three-dimensional task template

Every DR-ARENA-DEEP task composes three orthogonal data sources of the evaluation sandbox (Table 1):

Table 1: Three-dimensional deep-tier task template.

Dim.	Source	URL pattern	Inventory
A. Product landscape	Magento storefront	:7770/<slug>.html	~2 000 SKUs
B. Community sentiment	Postmill (reddit-like)	:9999/f/<forum>/<id>/<slug>	~30+ subforums
C. Technical grounding	Kiwix Wikipedia	:8090/content/<zim>/A/<title>	~6M articles

The task prompt requires an agent to triangulate claims across the three dimensions, producing a report that includes a cross-source synthesis section. A typical shallow-tier cross-site task touches

one or two dimensions and 5–20 URLs; a DR-ARENA-DEEP task touches three dimensions and 120+ URLs.

3.3 Task generation protocol

For each topic T we declare four lists in YAML (`configs/deep_topics/<id>.yaml`): *shopping keywords*, *reddit forums*, *reddit keywords*, *wiki mandatory*, *wiki extra*. The scraper `scripts/build_deep_golden.py` enumerates the sandbox programmatically, extracting product metadata, thread scores, and Wikipedia first-paragraph definitions. Empirically this yields ≈ 700 –800 pool URLs, ≥ 120 must-cite URLs, and ≥ 500 fact triples per topic in 30–45 minutes of sandbox-local traffic.

The resulting golden JSON is version-controlled; its hash is recorded alongside each task spec. Sandbox containers are pinned to specific image digests; no external network is contacted during an agent run.

3.4 Initial release (v1)

The v1 release consists of 10 topics: `audio_headphones`, `coffee_brewing`, `home_fitness`, `photography_starter`, `gardening_indoor`, `kitchen_cookware`, `pet_dog_supplies`, `baby_essentials`, `ev_roadtrip`, `mechanical_keyboards`. All are consumer-goods / consumer-technology domains; domain-diversity (biomedical, legal, policy) is planned for v2.

4 Evaluation framework

4.1 URL reachability verifier

Given an agent’s markdown report, we extract every `[label](url)` and bare-URL citation, canonicalise to `scheme://host:port/path?qs`, and restrict to the sandbox domains. Each canonical URL is probed with a GET, retried up to three times under exponential back-off with a concurrency cap of four to avoid self-DoSing the sandbox. We report

$$\text{reach} = \frac{|\{u : \text{HTTP}(u) = 200\}|}{|U_{\text{cited, sandbox}}|}.$$

This metric is deterministic and requires no gold reference, only a running sandbox.

4.2 URL coverage verifier

Given a curated must-cite pool M with per-URL weights w_m and a broader expected pool $P \supseteq M$,

$$\text{recall}_M = \frac{\sum_{u \in M \cap C} w_u}{\sum_{u \in M} w_u}, \quad \text{cov}_P = \frac{|C \cap P|}{|P|},$$

where C is the set of cited URLs. A domain-balance term $\min_d \frac{|C_d|}{\min_d |C_d|}$ enforces citations across shopping, reddit, and wiki. The URL-coverage score is $0.55 \cdot \text{recall}_M + 0.15 \cdot \text{cov}_P + 0.30 \cdot \text{balance}$.

4.3 Quote-match verifier (Layer 2)

Reachability catches *fabrication* (the URL does not exist) but not *misattribution* (the URL exists but does not contain the claim). For misattribution we extract a ~ 200 -character claim context window

around each markdown link, fetch the cited page, strip boilerplate, and compute a token-Jaccard between the claim window and the page text:

$$\text{quote_match}(c, u) = \mathbb{I}\left[\frac{|T(c) \cap T(u)|}{|T(c)|} \geq \tau_{\text{Jac}}\right],$$

with $\tau_{\text{Jac}} = 0.10$ (chosen empirically; below it any agent’s prose touches almost any page by coincidence). The score is the fraction of cited URLs passing the gate. This layer is deterministic, requires no LLM, and runs on the running sandbox.

4.4 Claim-NLI verifier (Layer 3)

The strongest available citation-verification protocol in the literature is NLI between the claim and a source quoted span [Gao et al., 2022, Anonymous, 2025b, 2024e]. For each cited URL we look up the golden’s `quoted_span` field — a ≤ 200 -character verbatim substring of the page that supports the canonical fact for that URL. We feed $(\text{claim}, \text{quoted_span})$ to DeepSeek-V4-flash in non-reasoning mode with an entailment-judge system prompt; the model returns a JSON object with an entailment probability p . We follow ReClaim [Anonymous, 2024e] in fixing $\theta = 0.80$ as the support threshold. The score is the fraction of evaluated claims with $p \geq \theta$. We cap the call budget at 60 NLI calls per report; results are stable beyond 40 in our pilot.

Quoted-span construction. The golden file’s quoted spans are LLM-drafted (DeepSeek-V4-flash; same model as the NLI judge but different prompt) and stamped `needs_human_review = true`. Per the methodology in our companion document, every span requires a human pass before publication; in the v0 audit reported here, spans are draft-only.

4.5 LLM-judge checklist

For each task we write 21 binary PASS/FAIL/UNCLEAR items covering dimensional completeness, synthesis findings, and citation format. A cross-family LLM judge (GPT-5-chat via a proxy) emits 21 verdicts, one line each; we report PASS rate.

4.6 Markdown-spec checks

Deterministic word count, citation count, and paragraph count ranges against the task-level `markdown_spec`.

4.7 Truthfulness-first composite (3-layer)

$$\text{quality} = 0.40 \cdot \text{url_cov} + 0.40 \cdot \text{judge_pass} + 0.20 \cdot \text{spec_pass}, \quad (1)$$

$$\text{truth} = \text{reach} \cdot (0.5 + 0.5 \cdot \text{qm}) \cdot (0.5 + 0.5 \cdot \text{nli}), \quad (2)$$

$$\boxed{\text{composite}_{v2} = \text{truth} \cdot \text{quality}} \quad (3)$$

Reachability is the only hard gate. If an agent’s reach is 0, its composite is 0 regardless of fluency. Layers 2 and 3 are soft gates of the form $0.5 + 0.5 \cdot m$, so a verifier returning 0 caps the contribution at 50% rather than annihilating it; this prevents small annotation errors in the golden quoted spans from killing otherwise-honest agents.

5 Experiments

5.1 Setup

All agents share a single backbone — DeepSeek-V4-flash in non-reasoning mode — served via a purpose-built OpenAI-compatible proxy that (i) injects `thinking:{"type":"disabled"}` so chat completions return content rather than `reasoning_content`, (ii) transparently rewrites `response_format=json_schema` (unsupported by DeepSeek) into `json_object` plus a schema-describing system nudge, and (iii) forwards `/v1/embeddings` requests to DashScope `text-embedding-v4` because DeepSeek has no embedding endpoint. The sandbox is Magento (port 7770), Postmill (port 9999), and Kiwix-English-Wikipedia (port 8090), each in pinned Docker images. A unified Tavily-compatible search shim proxies queries to all three.

5.2 Agents evaluated

- **GPT-Researcher** (Tavily-backed, graph-based) — v2 and v3, v3 with our DashScope embeddings route enabled.
- **LangChain** `open_deep_research` (LangGraph supervisor → researcher → writer).
- **HuggingFace** `smolagents` in code-as-action mode.
- **CAMEL-AI** ChatAgent with search toolkit.
- **Stanford STORM** (wiki-style multi-perspective).

5.3 Task

For the preliminary audit we run a single task `dr_cross_deep-0001` (audio headphones market intelligence; 125 must-cite / 739 expected pool / 559 triples). Scaling to all 10 topics and statistical CIs is in progress.

6 Results

6.1 Leaderboard

Table 2: Truthfulness-first leaderboard on `dr_cross_deep-0001` (single task, 5 unique agents, 6 runs). `Compositev2` uses 3-layer truthfulness gating per Eq. 3. *Reach.* = HTTP-200 rate of cited URLs (Layer 1); *QM* = quote-match (Layer 2); *NLI* = entailment rate at $\theta = 0.80$ (Layer 3, n in parens is the number of claims that had a golden quoted span available); *Judge* = GPT-5-chat checklist PASS count out of 21.

#	Agent	Reach.	QM	NLI (n)	Judge	Comp. _{v2}	Comp. _{v1}	Legacy
1	CAMEL-AI	0.98	0.95	0.00 (4)	5/21	0.139	0.284	0.293
2	smolagents	0.37	0.31	0.00 (7)	11/21	0.049	0.150	0.398
3	STORM	1.00	0.94	0.00 (1)	5/21	0.047	0.097	0.085
4	LangChain-ODR	0.03	0.00	0.00 (0)	12/21	0.003	0.013	0.320
5	GPT-Researcher (v2)	0.03	0.02	0.00 (0)	12/21	0.003	0.012	0.400
6	GPT-Researcher (v3, +embed)	0.03	0.01	0.00 (0)	8/21	0.003	0.010	0.333

6.2 F5 — LLM judge cannot detect URL fabrication

For GPT-Researcher (v2, v3) and LangChain `open_deep_research` 0–3% of sampled cited URLs resolve on the sandbox. These are *provably fabricated*: their URL paths (`/products/anker-soundcore-life-p2`, `/wiki/Bluetooth`) do not match any path the sandbox can serve. GPT-5-chat rates all three reports 21/21 PASS on the checklist.

6.3 F6 — Quantity/truthfulness ranking inversion

Table 3: Ranking inversion. Agents ordered by legacy composite (Eq. 4); the truthfulness-first composite (Eq. 3) reorders them in inverse. Δ is legacy rank – new rank; negative = truthful agent penalised under legacy, positive = fabricator rewarded.

Agent	Reach	Judge	Legacy	Legacy rank	New rank	Δ
GPT-Researcher (v2)	0.03	12/21	0.400	1	5	-4
smolagents	0.37	11/21	0.398	2	2	+0
GPT-Researcher (v3, +embed)	0.03	8/21	0.333	3	6	-3
LangChain-ODR	0.03	12/21	0.320	4	4	+0
CAMEL-AI	0.98	5/21	0.293	5	1	+4
STORM	1.00	5/21	0.085	6	3	+3

The legacy composite (Eq. 4) places the three *highest-fabrication* agents in first place; the truthfulness-first composite (Eq. 3) places them last. The ranking is not a perturbation — it is a reversal.

$$\text{composite}_{\text{legacy}} = 0.50 \cdot \text{url_cov} + 0.35 \cdot \text{judge_pass} + 0.15 \cdot \text{spec_pass}. \quad (4)$$

7 Discussion

Why do agents fabricate? The DeepSeek-V4 backbone is a high-capability frontier model with known tool-use hiccups on our proxy: (a) it lacks a JSON-schema mode, forcing LangChain to fall back to prompt-based JSON with lower reliability; (b) the instinct to produce “well-formatted” output dominates over faithful citation when the underlying framework does not enforce a grounding loop. GPT-Researcher recovered *zero* additional reachable URLs when we plumbed in an embeddings endpoint (DashScope v4), indicating that fabrication is *not* a tool-chain artefact but a policy artefact: the model receives real URLs from the shim and *chooses* to emit its own.

STORM and CAMEL-AI succeed by different routes. STORM’s multi-perspective conversational synthesis imposes a hard grounding loop: each claim is tied to a retrieved snippet. CAMEL-AI’s single- call ChatAgent succeeds by having fewer moving parts — there is no writer stage that could re-write away a real URL.

Why the legacy composite fails. The legacy composite is additive; a high judge score compensates for a zero URL-truth score. The truthfulness-first composite is multiplicative; there is no compensation. This is the exact structural change that flips the ranking.

8 Limitations

Single task. The results in this draft use `dr_cross_deep_0001` only. Statistical significance on the 10-topic v1 release requires 50 agent runs (≈ 4 h wall, \$3–5 API); that sweep is in progress. We expect both findings to strengthen rather than weaken with larger n .

Single backbone. All agents use DeepSeek-V4-flash so that agent architecture (not model choice) is the independent variable. A backbone ablation with GLM-4.5 and GPT-5-chat is planned.

Domain bias. v1 is entirely consumer goods. A biomedical / legal / policy deep tier (v2) is under construction.

Self-consistency of the judge. GPT-5-chat is a black-box hosted behind a third-party proxy; the model id returned in the response header has drifted during the study. Results are stamped with `judge_identity()` at evaluation time.

9 Conclusion

Short cross-site tasks and LLM-only judges measure surface fluency. Once tasks require 100+ URL citations on a reproducible sandbox, three out of five popular open-source Deep-Research frameworks are revealed to fabricate nearly every URL they cite, while conventional metrics rank them at the top of the leaderboard. Anchoring the composite score in HTTP-level reachability yields an opposite and far more honest ranking. We release DR-ARENA-DEEP, the verifier suite, and all agent outputs so this check can become a standard piece of any DR benchmark.

References

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A Reproducibility checklist

- Sandbox containers are pinned: `webarena_shopping` (magento-based), `webarena_reddit` (post-mill), `kiwix` (wikipedia_en_all_nopic). Image digests in `HANDOFF.md`.
- Backbone: `deepseek-v4-flash` via `api.deepseek.com` with `thinking:{"type":"disabled"}`.
- Judge: `gpt-5-chat-latest` via a third-party proxy (details in `.env`); judge identity stamped into every score file.
- Scripts: `scripts/build_deep_golden.py` (scraper), `scripts/run_deep_task.py` (agent runner), `scripts/score_deep_answer.py` (scorer), `scripts/score_deep_batch.py` (leaderboard).